Probing Novice Programmers' Everyday Interests to Inform Context-Personalized Programming Examples

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Motivation & Research Questions

Students in introductory CS courses often struggle to connect abstract programming concepts to their lived experiences. **Context personalization** adapts learning materials to incorporate students' interests, yet instructors lack systematic guidance on *which* everyday activities best illustrate specific computational thinking (CT) concepts.

Research Questions

RQ1: What CT concepts do students naturally associate with the ways they engage in their personal interests?

RQ2: Which types of engagement (e.g., play, create, consume) most frequently evoke CT concepts when students pursue their personal interests?

RQ3: Do different subpopulations pursue a different set of interests, or think differently about their CT approaches within those interests?

Method

Participants

278 undergraduates in CS1 courses at three U.S. universities (59.7% male, 36.7% female; 34.9% CS-related majors, 53.6% other majors; 34.9% first-generation college students, 65.1% non-first-generation students).

Procedure

Students listed up to 5 out-of-class activities and indicated which of 9 CT concepts they experienced in each activity, then described their two most vivid examples.

	Everyday Language	CT Concept
CT1	Do something over and over again	Count-controlled loop
CT2	Do the same set of tasks again	Functions
CT3	Remember something important	Variables
CT4	Keep doing until something changes	Condition-ctrl. loop
CT5	Make a mistake and investigate to find the error	Debugging
CT6	Make a choice based on context	Conditionals
CT7	Make a list	Arrays
CT8	Decide between two clear options	Boolean logic
CT9	Use a spreadsheet	Matrices

Table 1. Nine CT concepts investigated in the survey.

Analysis

Mixed-methods: chi-square tests, Cramér's V, thematic coding of free-text responses. Activities coded into **31 activity codes** grouped under **8 themes** (Play, Create, Consume, Connect, Explore, Nature, Work, Sleep).

RQ1: CT Concept Recognition Patterns

Three concepts dominate student responses (**54.19%** of all mentions):

- **CT6 — Conditionals**: 18.78% of mentions
- **CT3 — Variables**: 17.94% of mentions
- **CT1 — Count-controlled loops**: 17.46% of mentions

Chi-square goodness-of-fit tests rejected uniform distributions for 5 of 7 activity themes ($p < .001$), confirming that certain contexts naturally cue specific CT concepts.

RQ2: Activity Landscape

Play (43.60%) and **Create** (24.49%) together account for nearly **70%** of all activity–CT concept connections, providing the most generative contexts for personalization.

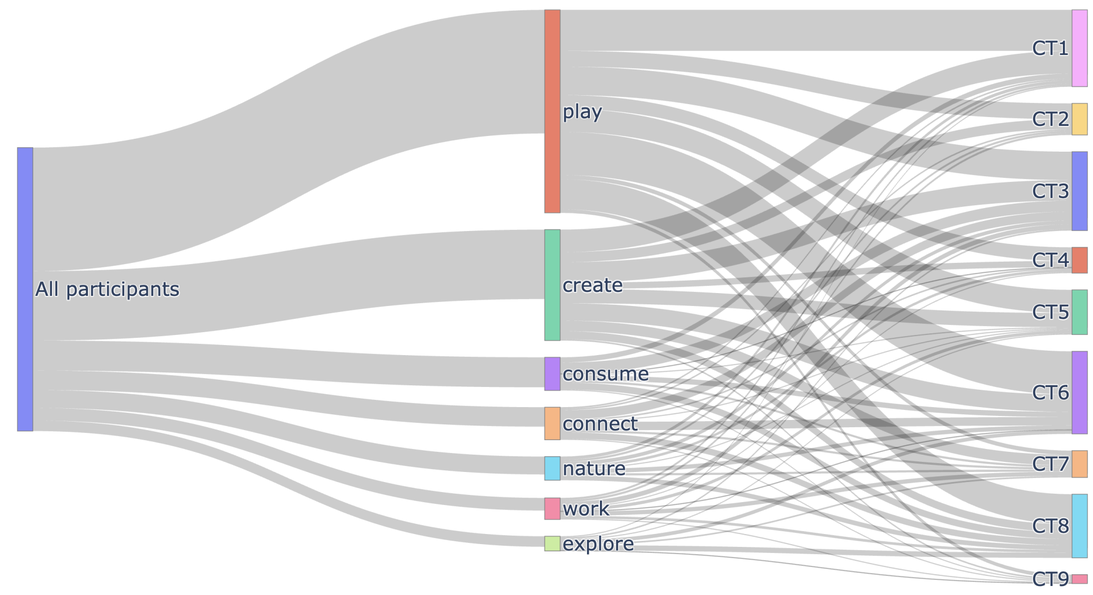

Figure 1. Sankey diagram: flows from activity themes to CT concepts (CT1–CT9).

RQ3: Demographic Differences

No statistically significant differences emerged at the overall, theme, or activity code level across three demographic dimensions:

Comparison	$\chi^2(8)$	p	V_{bc}
Gender (Male vs. Female)	8.84	.356	.032
Major (CS vs. Non-CS)	9.94	.270	.051
First-Gen vs. Non-First-Gen	6.46	.596	.000

Table 2. Overall tests of independence for CT concept mix (CT1–CT9).

⇒ **A common set of interest-based examples can serve diverse CS1 populations** without extensive demographic-specific customization.

Optimizing Example Sets: Five Strategic Subthemes

We refine *play* and *create* into **five subthemes**:

- **Gross Motor Play** — sports, gym, racing
- **Fine Motor Play** — video games, board games, puzzles
- **Static Creation** — art, writing
- **Dynamic Creation** — dance, music
- **Culinary Creation** — cooking

	play		create		
	gross motor	fine motor	static creation	dynamic creation	culinary creation
CT1	49	29	18	19	6
CT2	22	8	5	10	4
CT3	26	28	8	16	15
CT4	13	13	3	5	4
CT5	26	18	12	5	10
CT6	45	37	12	8	13
CT7	7	1	6	2	12
CT8	26	31	8	4	3
CT9	3	4	0	2	1

Figure 2. Refined subthemes × CT concepts. Darker = stronger association.

One example per subtheme per CT concept matches **57.1–84.2%** of students' stated interests (mean = **70.6%**).

RQ1 continued: Activity – CT Concept Frequency Distribution

	play				create					connect				consume		work			explore					nature						
	gym	play games	play sports	race	arts	cook	dance	music	writing	socializing	volunteer	pet	self-care	read	screen media	clean	learn	work	collect	driving	food	shop	travel	camping	fishing	garden	hiking	hunting	outside	walking
CT1	18	22	31	7	15	6	4	15	3	2	1	0	0	8	2	1	2	1	0	1	1	0	0	1	1	3	1	0	0	0
CT2	11	8	11	0	5	4	2	8	0	1	2	0	0	0	0	0	3	1	0	0	0	0	0	1	1	0	1	0	1	0
CT3	5	25	21	3	6	15	3	13	2	16	1	0	0	12	9	0	6	3	1	1	0	0	1	2	0	1	1	1	0	2
CT4	4	10	9	3	3	4	1	4	0	2	0	1	0	1	0	0	1	1	0	0	0	0	0	1	2	0	0	0	1	1
CT5	7	18	19	0	12	10	1	4	0	0	1	0	1	2	0	0	5	1	0	1	0	0	1	0	0	1	0	0	1	0
CT6	6	35	39	2	10	13	0	8	2	14	1	1	0	6	4	0	0	1	0	3	3	1	0	1	1	0	2	1	1	2
CT7	6	1	1	0	2	12	0	2	4	4	0	0	0	1	3	0	7	1	0	0	1	1	1	2	0	1	0	0	0	1
CT8	3	31	23	0	8	3	0	4	0	10	1	0	1	4	9	0	4	1	1	0	6	2	1	0	0	1	3	2	1	2
CT9	2	4	1	0	0	1	0	2	0	1	1	0	0	0	2	0	2	0	0	0	0	1	0	0	0	0	0	0	0	0

Figure 3. Theme × CT concept heatmap (counts). Darker cells indicate larger counts. Rows are CT concepts (CT1–CT9); columns are activity codes grouped by theme.

The heatmap presents the complete frequency counts of CT concepts across all 30 activity codes grouped by 7 themes. Play related activities–especially *play sports*, *play games*, and *gym*–show the strongest associations with CT6 (conditionals), CT1 (loops), and CT8 (boolean). Within Create, *cooking* uniquely highlights CT7 (arrays) and CT3 (variables), while *arts* and *music* connect broadly to CT1 and CT6.

Keywords: CS Education, Computational Thinking, Context Personalization, CS1, Situational Interest